

Hardware Issues

Q1: What is the maximum angle of joint 1 and joint 6 of myCobot 320 PI?

A1:

joint	mino	maxo	max speedo /s	max accelerationo/s2
J1	-165	165	150	200
J2	-165	165	150	200
J3	-165	165	150	200
J4	-165	165	150	200
J5	-165	165	150	200
J6	-175	175	150	200

Q2: How are the six servos controlled?

A2: The servo is controlled by the atom on the top.

Q3: What is the role of atom in the robotic arm?

A3: It controls the robot through algorithms such as forward kinematics, inverse kinematics, and coordinate switching. Atom is not open source yet.

Communication issues

**Q1: Why is there no display on the screen when the robot is connected with an HDMI cable?
Do I need to download the port driver?**

A1: Check if the connection is correct and the power is turned on. Try to use another interface and insert the interface stably. No need to download the port driver.

Q2: What versions of communication interfaces do different robotic arms support?

A2: The microprocessor-based robotic arm supports socket communication TCP; the microcontroller-based robotic arm can communicate via USB to serial port.

Q3: What is the communication frequency of the robotic arm?

A3: 10-20Hz.

Question about parameters

Q1: What is the speed unit of the robot arm?

A2: 180°/s.

Hardware Problem Solutions

1. How to deal with the vibration of the robotic arm?

Step 1: Burn the latest version of ATOM through myStudio.

Step 2: Upgrade pymycobot. Press Win+R and enter cmd to enter the terminal. Enter `pip install pymycobot --upgrade --user` and press Enter.

Step 3: Visit [GitHub](#) and download pid_read_write.py. Reset the parameters of the steering engine according to the prompts. Then run the system again.

Note that the parameters are set according to the data given in the figure below.

```
25 # 参数对应地址 corresponding parameter
26 data_id = [21, 22, 23, 24, 26, 27]
27 # 修改后的参数 reset parameters
28 data      = [32, 8, 0, 0, 3, 3]
29
```

Parameter	Connotation	myCobot pro 320		
		J1、J2、J3	J2	J5-J6
21	location loop P ratio coefficient	32	32	25
22	location loop D differentia coefficient	8	8	25
23	location loop I integral coefficient	0	0	1
24	minimun starting force	0	0	0
26	clockwise insensitive area	3	3	3
27	anticlockwise insensitive area	3	3	3

About the end gripper

Q1: Can the adaptive gripper not be fully closed?

A1: There is a certain gap between the jaws, and they are not completely closed. You can adjust it by increasing the thickness of the gasket between them.

Q2: What is the communication of the adaptive gripper?

A2: It is ttl communication.

Q3: What communication does the mycobot320 terminal provide?

A3: The mycobot320 terminal provides a 485 communication interface.

Q4: How to fix the USB camera on the end of the robot arm?

A4: Flange fixing is required and can be purchased independently.

Other issues

Q1: Why does the motor power shut off during use?

A1: Because the motor temperature is too high to use. Please wait a few minutes before using it again.

Q2: Does the robotic arm support Android-based development?

A2: We do not support Android development. If you want to develop it yourself, we can provide you with the port protocol.